

Improper integrals – 4

Principal value, symmetry and regularisation

Homework

12 сентября 2026 года

Theoretical minimum. The principal value is a deliberate cancellation of the two sides of a singularity, and it exists exactly when the divergent parts are equal and opposite; for a simple pole with a C^1 numerator this is automatic. Regularisation by subtraction attaches a value to a divergent expression by pairing it with another divergent one — Frullani's integral and the integral representations of Euler's constant are the model examples. Symmetry arguments prove equalities but never convergence, and applying them to a divergent integral produces nonsense. Finally, a product of sinc factors keeps the value π/a_1 until the widths $a_2 + \dots + a_n$ exceed a_1 : the Fourier transform turns the product into a convolution of boxes, and the value is the height of that convolution at the origin.

1. Compute v. p. $\int_{-\infty}^{\infty} \frac{dx}{x^2-1}$ and v. p. $\int_0^{\infty} \frac{x^{s-1}}{1-x} dx = \pi \cot(\pi s)$ for $0 < s < 1$, the second one by a real change of variable.
2. Prove that v. p. $\int_{-\infty}^{\infty} f(x - \frac{a}{x}) dx = \int_{-\infty}^{\infty} f(u) du$ for $a > 0$ and every integrable f , and state the general form of this identity for $x - \sum_k \frac{a_k}{x-b_k}$ with $a_k > 0$. (The one-sided version of the substitution is in the theoretical minimum of Integrals – 7.)
3. Define the finite part of $\int_0^1 \frac{dx}{x^2}$ by subtracting the divergent term of $\int_{\varepsilon}^1$, prove that the definition is consistent for $\int_0^1 \frac{f(x)}{x^2} dx$ with $f \in C^1$, and compute the finite part of $\int_0^1 \frac{e^x}{x^2} dx$ up to an explicit convergent integral.
4. Compute $\int_0^{\infty} (\frac{\sin x}{x})^4 dx = \frac{\pi}{3}$, and explain both this value and the value $\frac{\pi}{2}$ of $\int_0^{\infty} (\frac{\sin x}{x})^2 dx$ (computed in Integrals – 3 and again in Complex analysis – 4) in terms of the convolution picture of Fact 1.

5. Prove that $\int_0^\infty \frac{\sin x}{x} dx$ coincides with its principal value at infinity, and give an example of a function for which the improper integral diverges while the principal value at infinity exists and is non-zero.
6. Prove Frullani's formula in the regularised form: if $\int_1^\infty \frac{f(x)}{x} dx$ converges and $f(0^+)$ exists, then $\int_0^\infty \frac{f(ax)-f(bx)}{x} dx = f(0^+) \ln \frac{b}{a}$, and check it on $f(x) = \frac{1}{1+x}$.
7. Prove that $\int_0^\infty \left(\frac{1}{e^x-1} - \frac{1}{x} + \frac{1}{2} \right) \frac{dx}{x}$ converges, and express its value through γ or explain why no elementary expression is to be expected.
8. Let f be continuous on $\mathbb{R}$ with $f(x) \rightarrow L_\pm$ as $x \rightarrow \pm\infty$. Prove that v. p. $\int_{-\infty}^\infty \left(f(x) - \frac{L_+ + L_-}{2} \right) dx$ need not exist, but that $\lim_{R \rightarrow \infty} \frac{1}{R} \int_{-R}^R f = \frac{L_+ + L_-}{2}$ always holds.
9. Show by a symmetry argument that $\int_0^\infty \frac{\ln x}{1+x^2} dx = 0$, and then explain what goes wrong if the same argument is applied to $\int_0^\infty \frac{\ln x}{1+x} dx$.
10. Investigate what happens to the Borwein product when the factor $2 \cos x$ is inserted: decide for which n the value of $\int_0^\infty 2 \cos x \prod_{k=0}^n \frac{\sin \frac{x}{2k+1}}{\frac{x}{2k+1}} dx$ is still $\frac{\pi}{2}$, and formulate the general criterion in the language of Fact 1.